EMPLOYEE SALARY MANAGEMENT SYSTEM

SOUTH EASTERN RAILWAY

By

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We also wish to thank our team members for their dedication, hard work, and collaboration. The shared effort and collective input from each member were crucial in overcoming challenges and achieving our objectives.

Instructor's Signature:

Miss Rohini Bose (SE/IT,IT CENTRE,SER,GRC)

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INTRODUCTION

The Employee Salary Management System is a sophisticated application designed to streamline and automate the process of managing employee salaries within an organization. By leveraging Oracle 11g as the backend database, the system ensures robust data management, security, and scalability. The choice of Oracle 11g allows for efficient handling of large volumes of data and complex queries, ensuring the reliability and performance of the system.

The frontend is developed using ASP.NET and VB.NET, offering a user-friendly interface that facilitates seamless interaction for both administrators and employees. ASP.NET provides a powerful framework for building dynamic web applications, while VB.NET enables rapid development and ease of maintenance. Together, they create an intuitive and responsive user experience.

The project is built on the Visual Studio 2013 (VS 2013) framework, which provides a comprehensive development environment with advanced tools and features. VS 2013 enhances the development process with integrated debugging, code analysis, and performance profiling, ensuring a high-quality and efficient development cycle.

By integrating these cutting-edge technologies, the Employee Salary Management System offers a comprehensive solution for handling salary calculations and disbursements. The system aims to enhance the efficiency and accuracy of salary management, profile management and report generation, and ensure compliance with regulatory requirements.

FEASIBILITY STUDY

The Employee Salary Management System aims to automate salary processes, manage employee profiles, and generate reports using Oracle 11g (backend) and ASP.NET with VB.NET (frontend), built on VS 2013.

Technical Feasibility:

- **Technologies:** Oracle 11g ensures robust data management. ASP.NET and VB.NET provide a dynamic interface. VS 2013 offers a solid development environment.
- **Expertise:** The team is skilled in Oracle, ASP.NET, and VB.NET.
- **Infrastructure:** Existing hardware and software capabilities suffice.

Operational Feasibility:

- **User Training:** Minimal due to intuitive interface.
- **Maintenance:** Regular updates for security and efficiency.
- **Scalability:** Designed to grow with the organization's needs.

Economic Feasibility:

- **Development Costs:** Within budget.
- **Operational Costs:** Lower than current manual processes.
- **ROI:** Positive within the first year due to cost savings and improved efficiency.

TECHNOLOGIES USED

Hardware Requirements:

Server:

- Processor: INTEL G1
- RAM: 32GB
- SERPAS is a Rack Server
- OS Version: Red Hat Linux
- Network: High-speed internet connection

Client Machines:

- Processor: Intel Core i5 or higher
- RAM: 8GB or higher
- Storage: 256GB SSD or higher
- Display: 1920x1080 resolution or higher

Software Requirements:

Backend:

- Oracle 11g Database Management System
- Oracle SQL Developer (for database development and management)

Frontend:

- ASP.NET Framework
- VB.NET Language
- Visual Studio 2013 (VS 2013) Integrated Development Environment (IDE)

Development and Testing:

- Microsoft Windows Server 2012 or higher (for server)
- Microsoft Windows 10 or higher (for client machines)
- IIS (Internet Information Services) for hosting ASP.NET applications

TERMINOLOGIES

Server:

- A server is a powerful computer designed to manage network resources, host applications, and provide services to client machines. It typically has high processing power, ample memory, and significant storage capacity to handle multiple tasks and users simultaneously.
- In the context of our project, the server will host the Oracle 11g database, run web applications, and manage user requests and data processing.

SQL (Structured Query Language):

- SQL is a standard programming language used for managing and manipulating relational databases. It allows users to query, insert, update, and delete data within a database.
- SQL is used to interact with the Oracle 11g database, execute queries, and perform data management tasks such as retrieving employee records, updating salaries, and generating reports.

Database:

- A database is an organized collection of structured information or data, typically stored electronically in a computer system. It is managed by a Database Management System (DBMS).
- In our project, the Oracle 11g database will store all the information related to employee salaries, profiles, and other relevant data.

Schema:

- A schema is a collection of database objects such as tables, views, indexes, and procedures, organized under a specific database user. It defines the structure and relationships of the data within the database.
- Purpose: The schema in Oracle 11g will define the structure of the database, including tables for employee information, salary details, and other related entities.

View:

- A view is a virtual table in a database that provides a way to access and manipulate data from one or more tables. It is defined by a SQL query.
- Views can simplify complex queries by presenting data in a readable format, often used for reporting and data retrieval.

Trigger:

- A trigger is a database object that automatically executes a specified action in response to certain events on a table or view, such as insertions, updates, or deletions.
- Triggers in Oracle 11g can enforce business rules, maintain data integrity, and automate tasks such as logging changes or updating related tables.

Sequence:

- A sequence is a database object that generates a sequence of unique numbers, typically used for creating unique identifiers for rows in a table.
- Sequences in Oracle 11g are often used to generate primary key values automatically.

Procedure:

- A procedure is a stored program that performs a specific task. It can be executed to perform a series of SQL operations and can be called from other SQL statements or applications.
- Procedures in Oracle 11g can encapsulate complex business logic, automate repetitive tasks, and improve performance by reducing the amount of code executed on the client side.

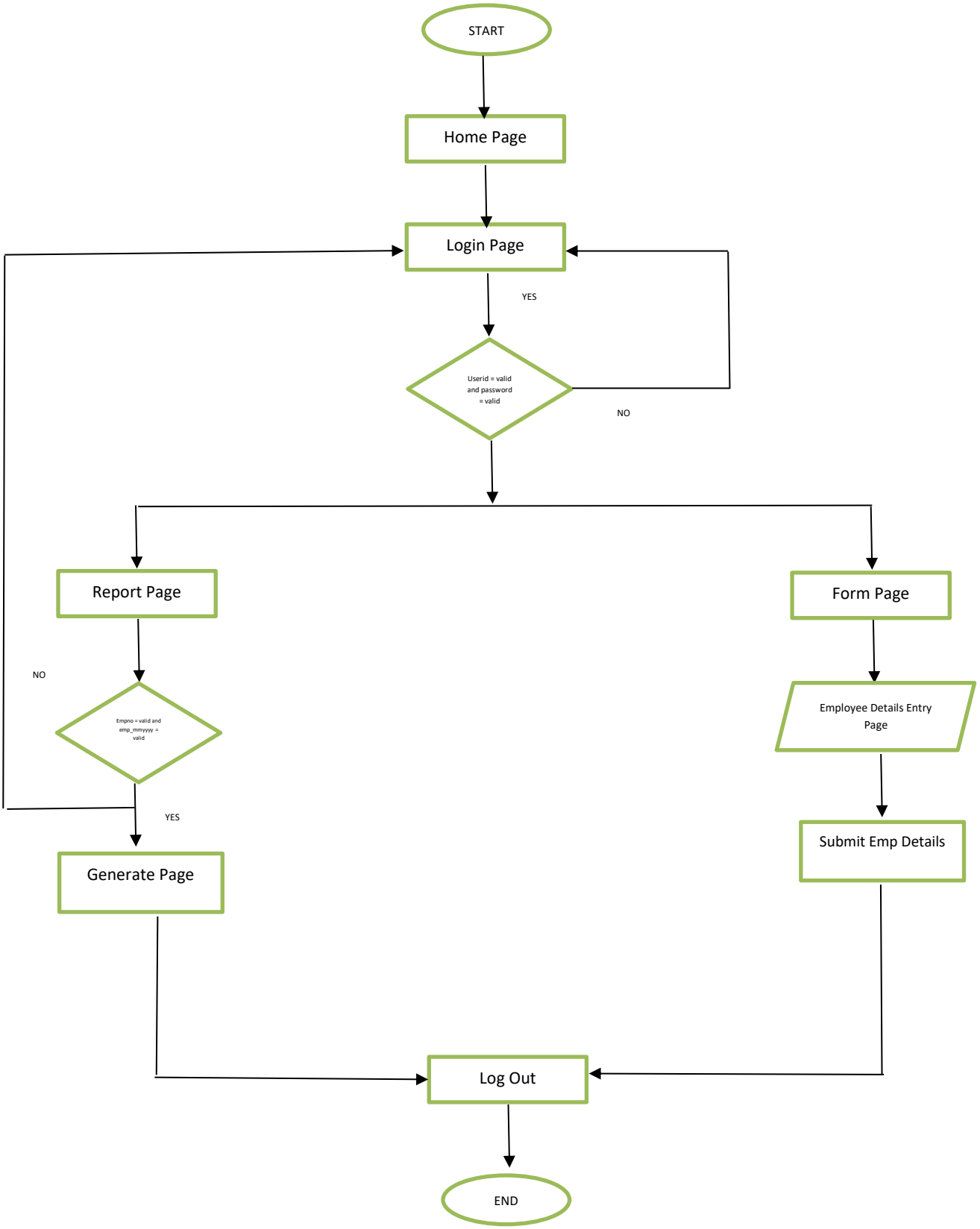
IIS (Internet Information Services):

- IIS is a web server developed by Microsoft for hosting and managing web applications and services. It supports various web technologies, including ASP.NET.
- IIS will be used to host our ASP.NET web application, providing the necessary infrastructure to serve web pages and handle HTTP requests.

Visual Studio 2013 (VS 2013):

- Visual Studio 2013 is an integrated development environment (IDE) from Microsoft. It provides tools and features for developing, debugging, and deploying applications.
- VS 2013 will be used to develop and manage the ASP.NET and VB.NET components of your Employee Salary Management System, offering a robust environment for coding, testing, and debugging.

FLOWCHART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Login Page
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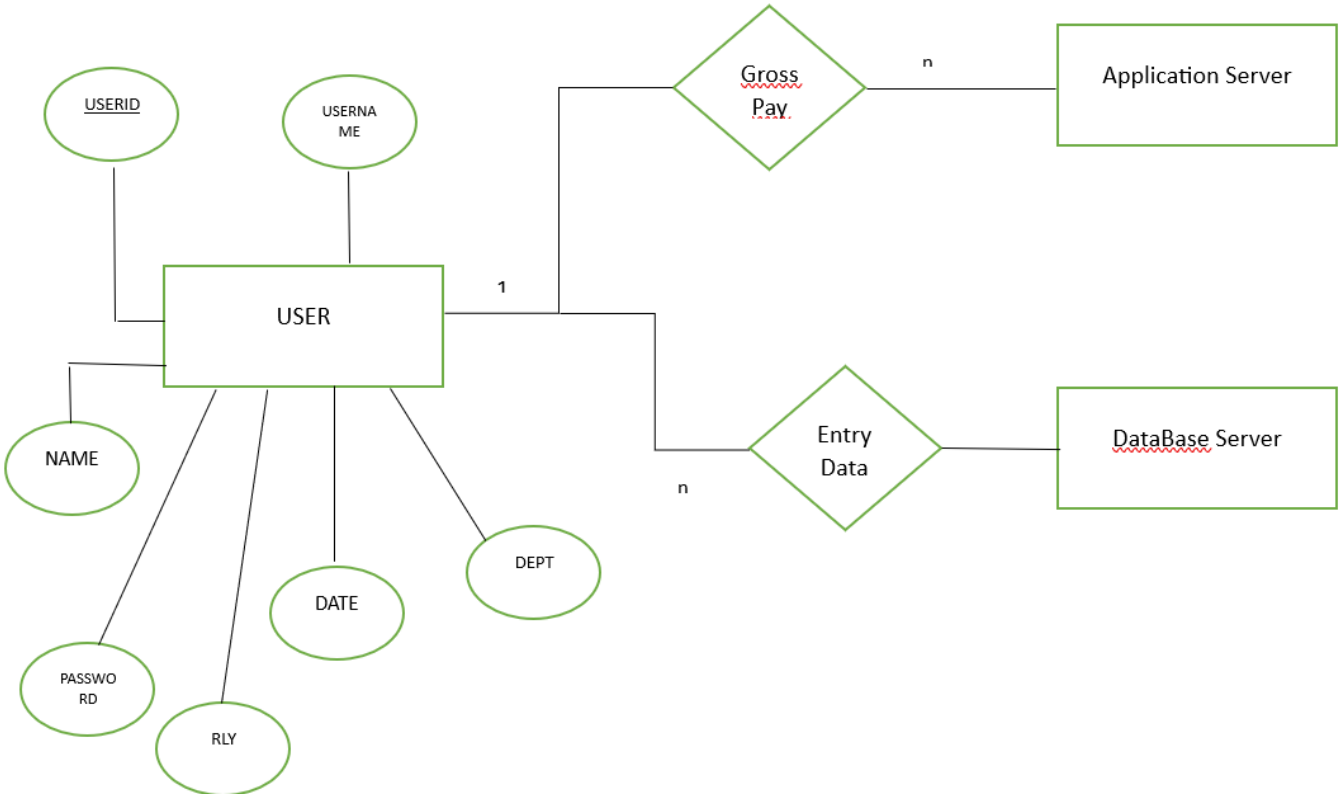
Userid = valid and password = valid
NO
Report Page
Form Page
NO
Empno = valid and emp_mmyyyy = valid
Employee Details Entry Page
YES
Submit Emp Details
Generate Page
Log Out
END

ER DIAGRAM


USERID
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Application Server
Entry
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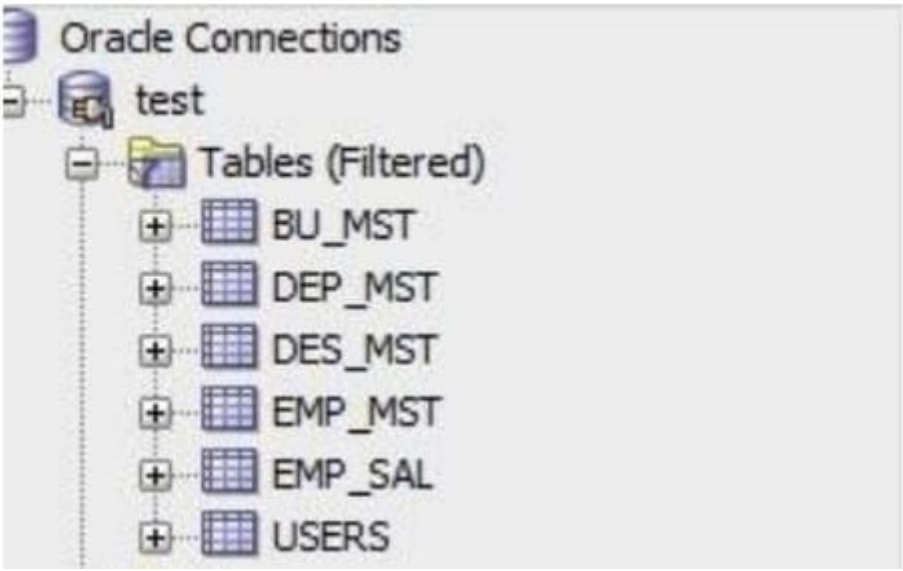
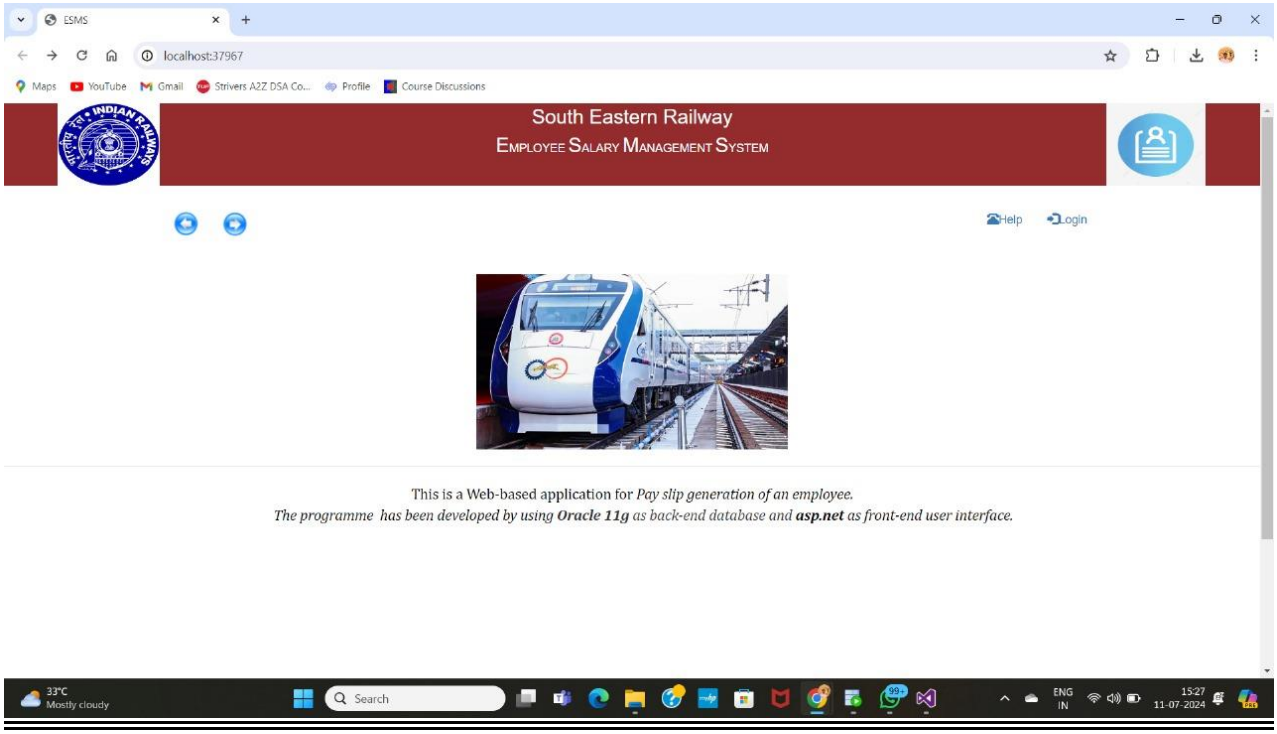
DataBase Server

TITLE SCREENSHOT



SCREENSHOT


ESMS
localhost:37967
Maps
YouTube
Gmail
Strivers A2Z DSA Co...
Profile
Course Discussions
South Eastern Railway
Employee Salary Management System
Help
Login
This is a Web-based application for Pay slip generation of an employee.
The programme has been developed by using Oracle 11g as back-end database and asp.net as front-end user interface.
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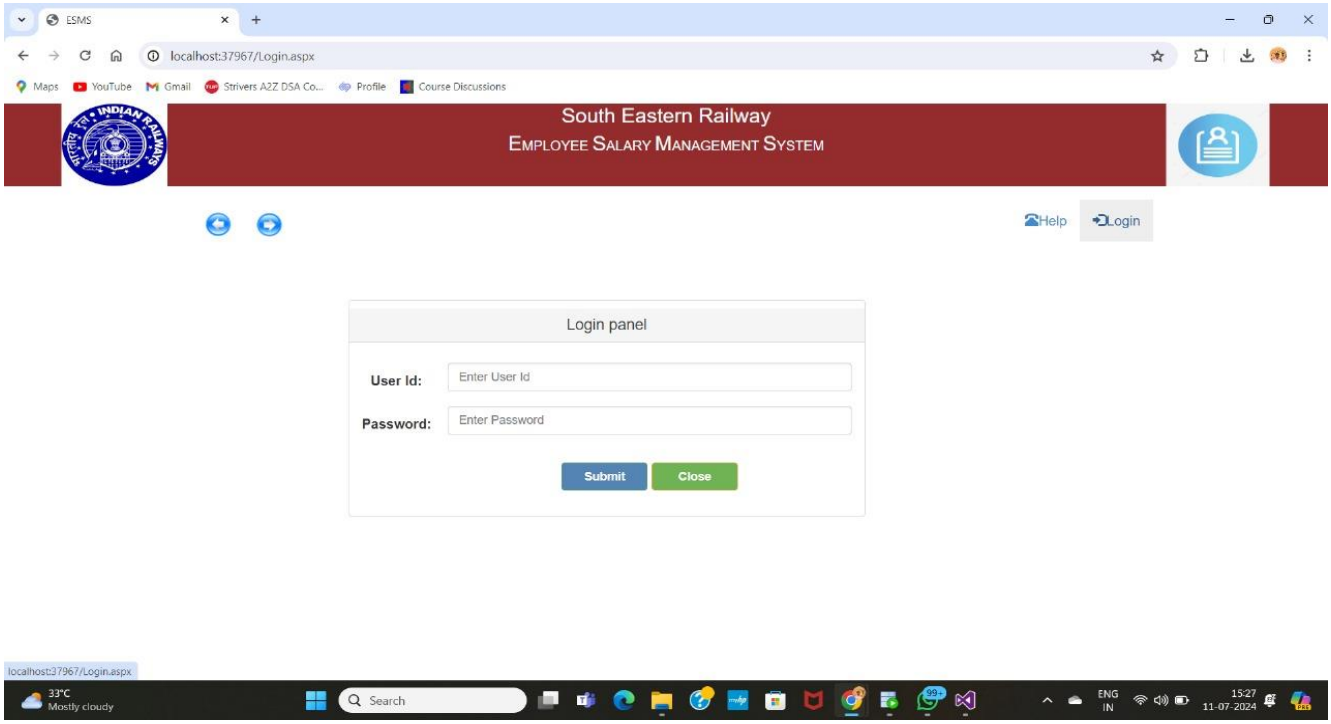
Search
ENG
IN
15:27
11-07-2024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Maps
YouTube
Gmail
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Profile
Course Discussions
South Eastern Railway
Employee Salary Management System
Help
Login
Login panel
User Id:
Enter User Id
Password:
Enter Password
Submit
Close
localhost:37967/Login.aspx
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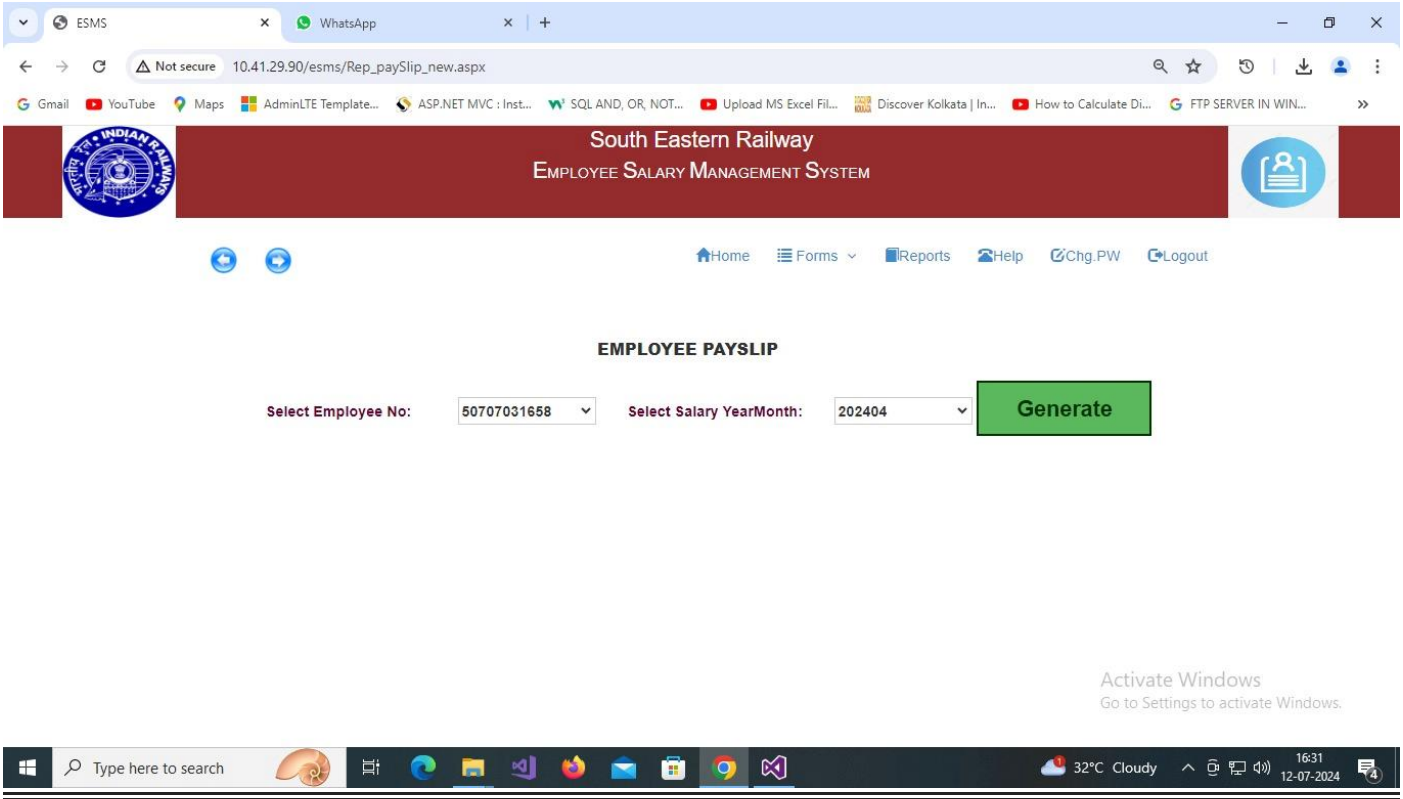
11-07-2024


ESMS
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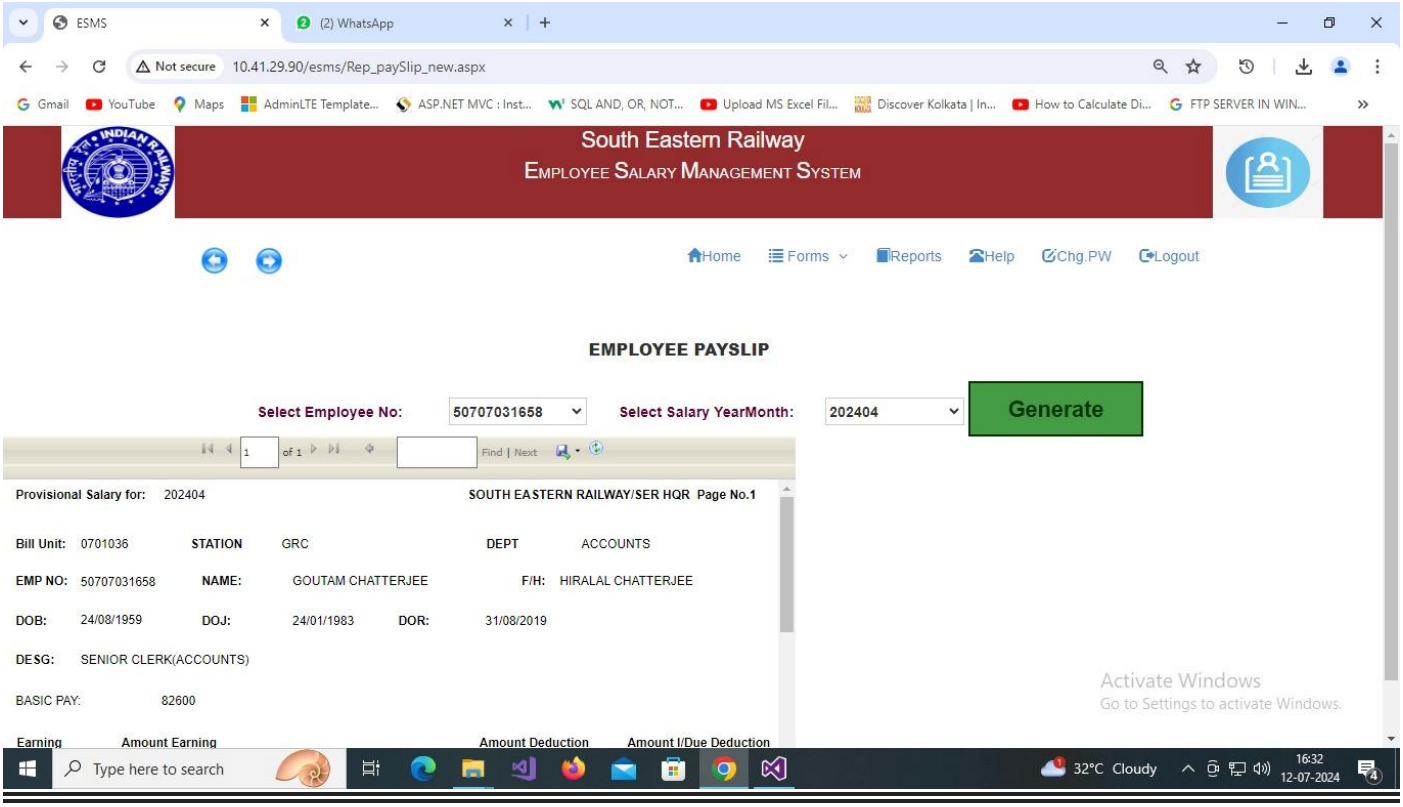
South Eastern Railway
EMPLOYEE SALARY MANAGEMENT SYSTEM
Home
Forms
Reports
Help
Chg.PW
Logout
EMPLOYEE PAYSLIP
Select Employee No:
50707031658
Select Salary YearMonth:
202404
Generate
Activate Windows
Go to Settings to activate Windows.
Type here to search
32°C Cloudy
16:31
12-07-2024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South Eastern Railway
EMPLOYEE SALARY MANAGEMENT SYSTEM
Home
Forms
Reports
Help
Chg.PW
Logout
EMPLOYEE PAYSLIP
Select Employee No:
50707031658
Select Salary YearMonth:
202404
Generate
Provisional Salary for: 202404
SOUTH EASTERN RAILWAY/SER HQR Page No.1
Bill Unit: 0701036
STATION GRC
DEPT ACCOUNTS
EMP NO: 50707031658
NAME: GOUTAM CHATTERJEE
F/H: HIRALAL CHATTERJEE
DOB: 24/08/1959
DOJ: 24/01/1983
DOR: 31/08/2019
DESG: SENIOR CLERK(ACCOUNTS)
BASIC PAY: 82600
Earning
Amount Earning
Amount Deduction
Amount I/Due Deduction
Activate Windows
Go to Settings to activate Windows.
Type here to search
32°C Cloudy
16:32
12-07-2024

Provisional Salary for: 202404 **SOUTH EASTERN RAILWAY/SER HQR Page No.1**

Bill Unit: 0701036 **STATION** GRC **DEPT** ACCOUNTS

EMP NO: 50707032067 **NAME:** RITA MONDAL **F/H:** LATE TARAPADA CHATTERJEE

DOB: 07/04/1962 **DOJ:** 24/01/1983 **DOR:** 30/04/2022

DESG: SENIOR CLERK(ACCOUNTS)

BASIC PAY: 87700

Earning	Amount Earning	Amount Deduction	Amount I/Due Deduction
PAY	87700	NPS	13155
DA	43850	WB TAX	200
HRA	23679	INCOME TAX	16062.90
TA	5400		

GROSS PAY: 160629 **DEDUCTION** 29417.90 **NET PAY:** 131211.10

BANK CODE: SBI **ACCOUNT NUMBER:** 37******891

FUTURE SCOPE

Integration with Other Systems:

- Integrate the salary management system with other HR systems, such as attendance tracking and performance management, to provide a comprehensive HR solution.
- Streamline HR processes, reduce data redundancy, and enhance data consistency across the organization.

Advanced Analytics and Reporting:

- Implement advanced analytics tools and enhanced reporting features to provide deeper insights into salary trends, employee performance, and financial planning.
- Support data-driven decision-making, identify cost-saving opportunities, and improve overall financial management.

Cloud Migration:

- Migrate the system to a cloud-based infrastructure to leverage scalability, flexibility, and remote accessibility.
- Reduce infrastructure costs, improve system reliability, and enable remote access for distributed teams.

Enhanced Security Features:

- **Description:** Implement advanced security measures such as multi-factor authentication, encryption, and regular security audits to protect sensitive employee data.
- **Benefits:** Ensure data security and compliance with regulatory standards, thereby enhancing trust and reliability.

AI and Machine Learning Integration:

- Integrate AI and machine learning algorithms to automate routine tasks, predict salary trends, and personalize employee experiences.
- Increase efficiency, reduce manual workload, and provide predictive insights for better HR management.

Employee Self-Service Portal:

- Create a self-service portal where employees can update their profiles, view salary slips, and request changes independently.
- Empower employees, reduce administrative workload, and improve user satisfaction.

CONCLUSION

The Employee Salary Management System represents a significant step forward in automating and optimizing payroll processes, profile management, and report generation within an organization. By leveraging advanced technologies such as Oracle 11g, ASP.NET, VB.NET, and Visual Studio 2013, the system ensures robust data management, user-friendly interfaces, and reliable performance.

The feasibility study has demonstrated the project's viability across technical, operational, economic, and schedule dimensions. The system is designed to enhance efficiency, accuracy, and scalability, thereby meeting the current and future needs of the organization.

Looking ahead, the system offers numerous opportunities for further enhancement. Integration with other HR systems, advanced analytics, cloud migration, and AI integration are just a few of the potential future developments that can expand the system's capabilities. Additionally, implementing enhanced security measures, customizable modules, and an employee self-service portal will further empower users and streamline administrative tasks.

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